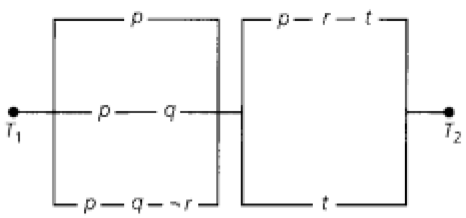
Internal Assessment Question Paper – 1

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of CSE

Programme: B. E | **Term: 2023-24(ODD)**
Course : Discrete Mathematical Structures | **CIE :** 1
Course Code : CS35 | **Sem:** 3 DIPLOMA
Section: A,B C | **Time: 3.30 -4.30 pm**
Date: 30-01-2024
Max Marks: 30 | **Portions for Test: L1-L10**

Instructions to Candidates: **<u>Question 1 is compulsory</u>**

1. Answer any one full question from 2 or 3. Each Question carries 15M.
2. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl#	Question	Marks	Bloom's Level	CO Mapping
1	a) Write the following in symbolic form and establish if the argument is valid: If A gets the supervisors position and works hard, then he will get a rise. If he gets a rise, then he will buy a new car. ∴ A did not get supervisors position or he did not work hard. (ii) State which rule of inference is the basis for the following arguments: a) If it snows, then school is closed. School is open. Therefore, it is not snowing. b) Alyssa is talented and smart. Therefore she is smart	5	Apply	CO1
	b) Define a partial order on a set with an example? Construct the Hasse Diagram for the divides relation on the following set A={2,4,6,9,12,18,27,36,48,60,72}	4	Apply	CO2
	c) Define open statement with an example. With the universe of students in a particular college and with the following open statements: j(x): x is a junior, s(x): x is a senior, p(x): x is enrolled in a physical education class, represent each of the following arguments in the symbolic form and prove the validity of the argument. No junior or senior is enrolled in a physical education class Mary Gusberti is enrolled in a physical education class. Therefore, Mary Gusberti not a senior	6	Apply	CO1
2	a) Simplify the following network using laws of logic: 	6	Apply	CO1
	b) Define an equivalence relation with an example. Let A = {1, 2, 3, 4, 5}, give an example of a mapping which is (i) neither symmetric nor antisymmetric (ii) anti-symmetric and reflexive but not transitive	4	Understand	CO2

	(iii) reflexive, transitive but not symmetric			
	c) With steps and reasons establish the validity of the argument: $\forall x[p(x) \vee q(x)]$ $\exists x \neg p(x)$ $\forall x[\neg q(x) \vee r(x)]$ $\forall x[s(x) \rightarrow \neg r(x)]$ ——————— $\therefore\ \exists x \neg\, s(x)$	5	Understand	CO2
3	a) Let A = {2, 3, 4, 6, 8, 12, 16}. aRb if and only if ‘a divides b’ (i) Write R (ii) Give matrix of R (iii) Draw digraph of R (iv)Indicate the properties of the relations it holds?	4	Understand	CO2
	b) Give a direct, indirect and contrapositive proof of the theorem for “If n is an odd integer, then n^2 is odd.”	5	Apply	CO1
	c) Prove the logical equivalence using laws of logic: (i) $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$ (ii) $(\neg p \wedge (\neg q \wedge r)) \vee ((q \wedge r) \vee (p \wedge r)) \Leftrightarrow r$	6	Apply	CO1

Course Outcomes to be accessed by the IA Test 1:

CO1: Apply the notion of mathematical logic & proofs in problem solving.
CO2: Solve problems which involve discrete data structures such as relations